

All of the above mentioned vendors provide excellent support in terms of software and all the above mentioned devices can be used with ease once you get acquainted with them.

2.5 Op-amp

Op-amp stands for operational amplifier. It is just like an amplifier which you use while listening to music. It performs the function of amplifying the voltage signal input to it, converting it into a voltage of significantly higher magnitude. Generally, it comes in form of an IC and it has multiple ports to take multiple signals and amplify them simultaneously. The most popular Op-amps among hobbyists are MCP series and LM series Op-amps. We are going to use LM358N Op-amp for our purpose.

2.6 Motor driver

A motor driver is an IC which is capable of providing voltages sufficient enough to drive D.C. motors. We use a L293D motor driver. The L293D is designed to provide bidirectional drive currents of up to

600mA at voltages from 4.5 V to 36 V. It can also drive inductive loads such as relays, solenoids, dc and bipolar stepping motors, as well as other high-current/high-voltage loads in positive-supply applications.

2.7 Sensors

Robots are very similar to human body. Human body has senses to receive information from the external environment. Robots have sensors. Human body has brain to process the information; robots have processors to do the same. Human body have the external organs to execute decisions given by mind, in robots the wheels or legs or arms does it. There are two broad categories of sensors:

Digital sensors: They provide discrete or stepped binary results. A switch an example of a digital sensor: It can either be open or closed. Also, an ultrasonic ranger, which returns a different binary value for distances, is also a type of digital sensor.

Analogue sensors: They provide a continuous range of values, usually in form of voltage.

Generally, in both digital and analogue sensors, the input to the processing unit is voltage. In the case of a digital sensor, the processing unit is interested in knowing whether the voltage input given by the sensor is a logic low (usually 0 V) or logic high (usually 5 V). Most of the digital sensors can be